

Investigation of a community-based clean energy system holistically with renewable and hydrogen energy options for better sustainable development

Muarij Khalil^{*}, Ibrahim Dincer

Clean Energy Research Laboratory (CERL), Faculty of Engineering and Applied Science, Ontario Tech University, 2000 Simcoe Street North, Oshawa, Ontario, L1H 7K4, Canada

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ABSTRACT

This study develops a novel community-based integrated energy system where hydrogen and a combination of renewable energy sources are considered uniquely for implementation. In this regard, three different communities situated in Kenya, the United States and Australia are studied to cover their needs potentially by producing hydrogen, biogas, power, residential space and industrial heating, hot water and fresh water. where a multi-generational geothermal plant integrated with a solar power plant (using both concentrated solar power (CSP) and photovoltaic (PV) panels) is considered to provide the energy inputs. Innovations in hydrogen production and renewable energy are essential for reducing carbon emissions. By combining the production of hydrogen with renewable energy sources, this system seeks to move away from the reliance on fossil fuels and toward sustainability. The study investigates various research subjects using a variety of methods. The performance of the geothermal source is considered through energetic and exergetic thermodynamic analysis. The software System Advisor Model (SAM) and RETScreen software packages are used to analyze the other sub-systems, including CSP and PV units and a combined heat and power plant. The Australian, American, and Kenyan communities considered for this study are found to have promising potential for producing hydrogen and electricity from renewable sources. The geothermal output is expected to be 35.83 MW for electrical power, 122.8 MW for space heating, 151.9 MW for industrial heating, and 64.25 MW for hot water. The overall multigenerational geothermal energy and exergy efficiencies are reported as 65.15% and 63.54% respectively. These selected communities are expected to have annual solar power generation and hydrogen production capacities of 158 MW, 237 MW, 186 MW, 235 tons, 216 tons, and 313 tons respectively.

1. Introduction

The need for sustainable energy solutions and renewable energy sources has drastically increased due to an increase in global initiatives to reduce carbon emissions (IRENA, 2023). Significant work still needs to be done to achieve net-zero carbon emissions by 2050. Sustainable solutions are required to meet current energy demands, such as for transportation fuels, heating, and electricity. Interestingly, there has been a significant shift in the last few years toward the use of renewable resources for transportation and electricity generation. A new approach to energy storage is to produce hydrogen using excess energy from renewable sources which can then be used as an alternative fuel. Hydrogen production through electrochemical water splitting with an electrolyser is becoming popular (EERE, 2023).

There have been challenges for effective fresh water, energy

production, wastewater treatment, and their effects on necessities (Elbashir, 2017). The need for sustainable energy solutions has been highlighted in many books, publications, and papers by many researchers. Basit et al. (2020) explore the problems and solutions related to the integration of renewable energy systems into the electrical grid. Their study investigates how energy storage systems improve system stability, efficiency, and energy density in renewable energy systems-based power systems. However, their study does not consider how renewable energy production and storage changes for multiple integrated renewable energies. Vuuren et al., 2012 discuss the numerous issues that face the energy sector, ranging from long-term sustainability and security to consumption and environmental concerns. Their paper explores a vision for the global energy framework, promoting coordinated approaches, increased energy efficiency, and large-scale investment scaling. Although, the study does not consider the specific

^{*} Corresponding author.

E-mail addresses: Muarij.Khalil@ontariotechu.ca (M. Khalil), Ibrahim.Dincer@ontariotechu.ca (I. Dincer).

challenges that would be faced by different communities and regions, which would be an important consideration for energy investments. Lazdins et al. (2021) conducted a thorough analysis of 64 recent publications from 2015 to 2021, their study explores the role that communities and individuals are playing in advancing solar energy systems to meet future energy and climate goals. Their paper further examines the advantages and obstacles to the establishment of solar energy communities, emphasizing the role that citizens play in implementing these systems. Suggested solutions for promoting PV solar technologies include promoting communities and satisfying consumer needs, the research examines policy, economics, technology, and social aspects, focusing on areas like policy frameworks, economic assessments, energy management, and consumer adoption. Although the study dives deeply into solar contribution, their study does not consider investments in other energy and renewable sources which could be a factor in prompting solar power. Further, Hoffman et al. (2022) emphasize that despite initiatives like the United Nations Framework Convention on Climate Change (UNFCCC), there are still difficulties in reducing greenhouse gas emissions. They discuss obstacles such as economic interests and political shortsightedness. The study emphasizes the need for cooperative and moral long-term policies and promotes other ways to lower global temperatures without going over crucial benchmarks.

Moreover, some studies have examined a combination of potential sustainable solutions for mitigating climate change. Khalil and Dincer (2023) considered a combination of renewable energy sources to investigate hydrogen production. Their study considers communities in South Africa, Denmark, the Netherlands, and Canada to evaluate their system and how to reduce carbon emissions. Their proposed system utilizes both onshore and offshore wind power and PV solar energies for hydrogen production, and for meeting the energy needs of the communities. A compressive analysis of the various available energy sources is considered, however, economical factors may pose challenges in implementing these sources. In a study, HassanzadehFard et al. (2020) integrated electrolyzers, hydrogen storage, and fuel cells, as well as optimizing microgrid operations with renewable energy sources like photovoltaic arrays and wind. In seasons with high renewable energy output, their study evaluates two hydrogen production techniques to increase efficiency and reduce methane consumption by 10%. Their study emphasizes how important electrolyser technology is in reducing methane consumption in microgrid systems. A possible area for further investigation could include considering scalability and the adaptability of the technology. Temiz and Dincer (2021) present a new system that effectively produces hydrogen by combining solar and nuclear energy sources where a high-temperature type advanced nuclear reactor and bifacial PV-based solar based integrated system is employed to meet the needs of energy, food and fuel, particularly designed for sustainable indigenous cities in Canada. Hydrogen is produced and stored by combining these energy sources with established water-splitting processes. Their study conducts a parametric evaluation of various system capacities and energy and exergy efficiencies under various operating conditions. An important consideration for their system not evaluated in their study is environmental impact assessment that will need to be addressed by combining solar and nuclear. Al-Shetwi (2022) examines current global trends, environmental impacts, and challenges to determine how the integration of renewable influences sustainable development. They examined 177 of the 712 critical papers published between 2010 and 2021, to highlight the advantages of integrating renewable energies while also addressing their drawbacks and new difficulties. Future research work related to their study is expected to focus on the role of social acceptance and how different global communities come into play in sustainable development.

Some studies have considered energy systems utilizing integrated renewable energy sources to provide multiple useful outputs. Bolt et al. (2021) developed a new renewable energy system that uses molten salts to store thermal energy while producing freshwater, hot water, heating, cooling, and electricity. Upon analyzing operations in different

scenarios, their system demonstrated an overall energetic efficiency of 54.42% and an exergetic efficiency of 30.24%. Energy related coefficient of performance for heating and cooling were 0.8169 and 2.265, respectively, whereas exergetic coefficient of performance were found to be 0.2683 and 0.3864, respectively. Ahmadi et al. (2022) present a multi-generation solar system that combines geothermal energy with parabolic trough solar collectors. It is thoroughly analyzed in three operating modes and shows significant gains in terms of effectiveness and heat dissipation utilization. Their system efficiently handles cooling loads and exhibits optimized production of hydrogen and fresh water. A potential gap that can be addressed in future work should consider the cost-effectiveness and economic analysis of implanting the system for the modes evaluated. Mohammadi et al. (2020) examine the possible configurations, benefits, difficulties, and possible future directions of solar energy-driven multigeneration systems through a thorough review and analysis. Their paper focuses on two primary groups, solar-only systems and hybrid solar-driven configurations that integrate solar power with additional renewable energy sources. The integrated systems are shown to provide decreased carbon emissions, increased efficiency, and lower costs. Potential climate changes due to weather fluctuations can affect the efficiency of the system which would be an important consideration for their study. Haider et al. (2023) looked at a multigeneration system with geothermal energy as its main source. Power, cooling, fresh and hot water, dry air, hydrogen, and space heating are among the outputs that their system can provide. With the use of double flash desalination, double effect absorption, and regenerative Rankine cycles, the system produces a significant amount of electricity, hydrogen, and fresh water fit for residential or commercial use. At 103 MW of output and 24.42% thermal efficiency is obtained. The system also reports 54.22% and 38.96% energetic and exergetic efficiencies, respectively. However, their study does not consider environmental conditions, water usage, and land impact based on location which will impact the outputs from the geothermal source.

Many researchers have found advantages for hydrogen production for energy storage and as an energy carrier. Lonis et al. (2020) investigated small-scale decentralized energy systems that use captured CO₂ and renewable hydrogen to produce methanol. It investigates two methods of electrolyzing water, examines the conversion procedures, and employs Aspen Plus simulations to assess the results. It is important to consider that the feasibility of the system may differ based on the technology available realistically. In another study, Kalinci et al. (2015) model a hybrid renewable energy system using hydrogen storage for Bozcaada Island in Turkey. By optimizing equipment sizes, and combining wind and PV solar power, their system shows to reduce costs significantly. However, their study does not discuss the long-term stability of their hybrid system. This will impact the reliability of the system for varying weather patterns. Aravindan et al. (2023) examine different non-renewable sources for hydrogen production methods and evaluate their efficiency, cost, and environmental impact. The study finds that improvements in technology are needed for greater efficiency and in expanding clean hydrogen generation for a widespread hydrogen economy, even though each method has advantages of its own. For example, nuclear-based production offers cleaner hydrogen but at higher costs, and coal gasification provides flexibility. Hren et al. (2023) assess the effects of various hydrogen production, storage, and transportation methods on the environment. The study finds that water electrolysis uses more energy than it generates. Their study finds that the most environmentally friendly way involves pipeline transportation of gaseous hydrogen, with an energy footprint of between 38% and 85%. These results point to less taxing methods for producing and moving hydrogen. Nasser et al. (2022) evaluate a hybrid renewable energy system that produces hydrogen in several Egyptian cities by utilizing photovoltaic panels and wind turbines. Significant electricity production is demonstrated by the results, allowing most cities to generate hydrogen. Cities produce different amounts of hydrogen based on the solar potential in each area. The production costs ranged from 4.54 \$/kg

to 7.48 \$/kg, and the system efficiency falls between 7.69% and 9.37%. The study does not discuss the cost needed to make hydrogen an acceptable source for different communities. Feng and Menezes (2022) present an inventory model that integrates storage and dual energy sources. The management of such complex energy mixes in conjunction with storage technologies. Using hydrogen as an energy-storage solution, they present a production and inventory model for periodic reviews. The various barriers posed for large-scale applications and in energy usage and storage are not considered in their study. These would be important factors for the feasibility of the system and the practicality of industrial applications.

Furthermore, numerous researchers have looked at various approaches for producing hydrogen. Parthasarathy and Narayanan (2014) conducted a review of hydrogen production from biomass gasification. Their review discusses the various parameters of the impact on hydrogen yield. Their study finds that biomass type, catalyst, and the ratio of steam to biomass ratio play an important role in the hydrogen yield. It is important to consider that the optimal biomass ratio as well as the different parameters would differ for the unique type of biomass used which would need to be accounted for. In particular, Ryczkowski et al. (2017) looked at lignocellulosic biomass (wood) for hydrogen production using Ni/CaO-ZrO₂ catalysts. Their study shows that the composition of the material and preparation influences the production. In another study, Sandeep and Dasappa (2014) performed a study using steam and oxygen to produce hydrogen-rich syngas from biomass gasification. They developed an experimental setup to look at different steam-to-biomass ratios for hydrogen yield. They found that the hydrogen yield increased at higher steam and biomass ratios. They found a peak hydrogen yield of 104 g/kg biomass at a steam and biomass ratio of 2.7. Although the hydrogen yields increased, the study showed that the efficiency of the system decreased at higher steam-to-biomass ratios.

Green hydrogen production has also commonly been considered using renewable energy sources such as wind and solar. Electrolysis is commonly used to produce green hydrogen by splitting water using excess renewable power. Liponi et al. (2023) looked at the feasibility of green hydrogen production using wind. Interestingly, the study considered separating the electrolysis into multiple groups to determine the profitability of their system. A 39 MW wind farm is used, and their study finds that dividing the plant into two separate groups results in higher hydrogen production and more profit. However, dividing the plant into greater than two groups didn't show to be beneficial. Their study only considers a fixed capacity, therefore for lower plant capacities using this approach to divide the plant into groups for hydrogen production may not be beneficial. On the other hand, considering other or hybrid renewable sources such as solar may also result in different types of configurations as more beneficial than the ones suggested by their study. Some researchers have also considered using solar power for hydrogen production. Shen et al. (2023) conducted an experimental investigation of hydrogen production and nanofluids-assisted thermal systems for a PEM electrolyser. Their study found that using nanofluids resulted in increased electrical and thermal efficiencies. Hence, their system also increased hydrogen yield, at 0.012 kg/s, and the thermal efficiency was reported as 33.4% with the highest hydrogen production ratio of 17.4 ml/min. In another study, Khelifaoui et al. (2021) conducted an experimental study for hydrogen production using solar, and PEM electrolyser. Algerian Sahara region is considered, and the results showed that a high hydrogen of 284 L was obtained from an 8-h operation in one day. The efficiency fluctuated between 18% and 40% during operation. To obtain a deeper understanding of their system, long-term operation should be considered on an annual basis to determine how effective the system operates during different weather conditions.

This study introduces a novel approach to meeting community energy needs without depending on fossil fuels. It deviates from traditional methods by concentrating on using solar and geothermal energy to produce electricity, heating, fresh water, hot water, biogas, and

hydrogen. A lack in the current body of literature is the investigation of solar and geothermal energies to produce multiple useful energy outputs. This study differs from previous ones in that it looks at these renewable sources in three different global communities that have never been investigated before. This research offers an unprecedented analysis of the potential of solar and geothermal resources, utilizing a combination of different clean energy conversion processes. This study is noteworthy not only for its emphasis on renewable energy compositions but also for its distinct approach to examining the potential of hydrogen use.

The main objective of this paper is to develop a sustainable energy system using a combination of renewables to meet the energy needs of a community with hydrogen production as an energy storage medium. The specific goals include analyzing system performance in different communities with different climates. The system is to be further assessed using performance metrics and efficiency. In addition, the study aims to determine the potential for meeting the electrical demands and estimate hydrogen production based on available renewable resources.

2. System description

The suggested renewable energy system, shown in Fig. 1, combines several elements to maximize resource usage and energy production. Photovoltaics (PV) and Concentrated Solar Power (CSP) are used in the hybrid solar system to produce high-temperature steam that is used to generate electricity. To facilitate steam biomass gasification for the production of hydrogen, this steam can also be directed toward the Hydrogen Generation Plant (HGP). The HGP uses alkaline electrolysis to produce hydrogen from excess renewable energy. In addition, if solar and geothermal energy sources are insufficient, the Combined Heat and Power (CHP) plant uses biogas produced from accessible biomass to continue generating electricity. Furthermore, the CHP plant effectively recovers heat to provide space heating for the.

Hydrogen production is essential as an energy carrier and storage medium. The HGP then uses solar heat, to produce hydrogen using biomass gasification. In addition, hydrogen is also produced using the alkaline electrolyser utilizing access renewable power from both geothermal and solar sources. The hydrogen produced has two uses, it can be stored as energy for use in the future when renewable energy output is low. Second, hydrogen can be used as an alternative fuel for fuel cell-based cars, providing a more environmentally friendly mode of transportation.

One crucial component incorporated into the system is clean water production. The Water Treatment Plant (WTP) draws water from lakes or the ocean (based on what is available locally), and then is distributed to the community for use. A steady supply of water is maintained through the storage of extra water in the community water storage tanks. Additionally, the HGP receives water from the WTP, which facilitates hydrogen production.

The hybrid solar plant and the geothermal system provide the energy that is distributed to the community and homes to provide household electrical needs. The multigenerational geothermal plant recovers heat in addition to producing electricity. The community effectively uses this recovered heat for space heating, industrial processes, and hot water supply, improving energy efficiency and satisfying a variety of other energy requirements.

2.1. Methodology

This study utilizes various research methods and software to analyze the proposed system. The geothermal and solar energy sources are further defined to identify the specific system under consideration. The subsystems of the solar concentrated plant and multigenerational geothermal plant are shown in detail in Fig. 2. The geothermal plant uses the heat from solar to create steam, which powers turbines to produce electricity. To further increase electricity output, the solar plant

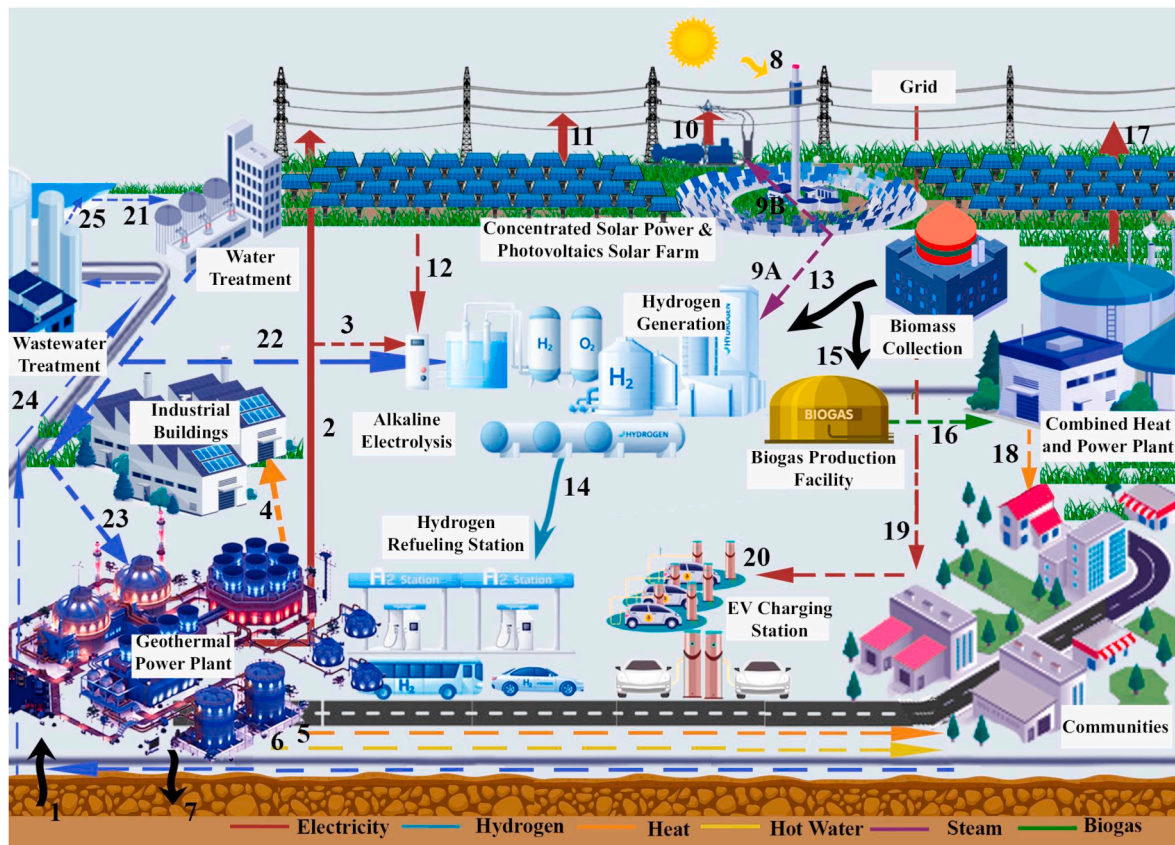


Fig. 1. The proposed hydrogen generation and renewable system.

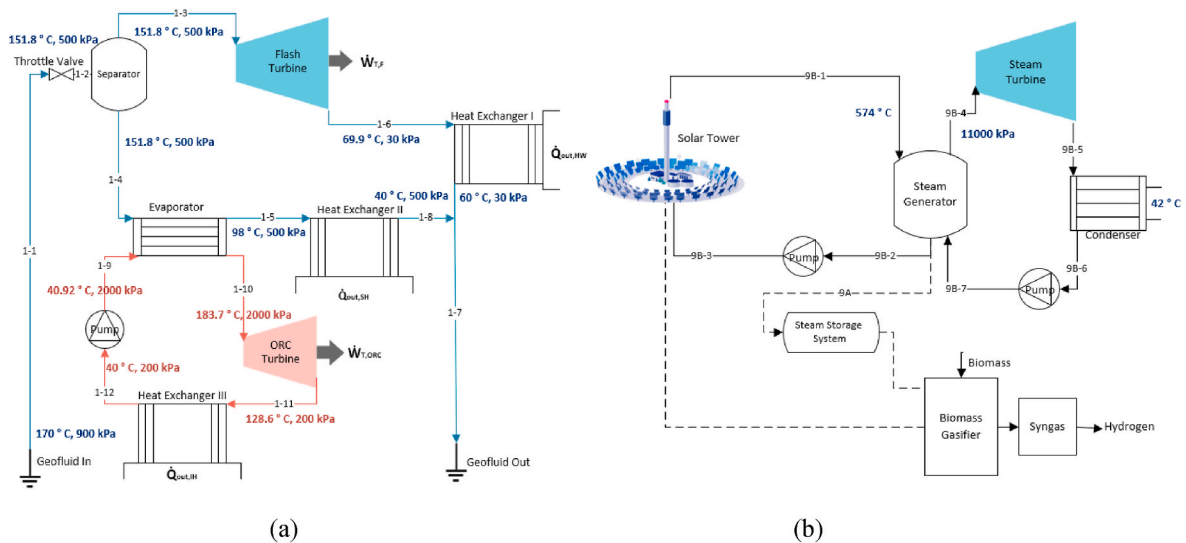


Fig. 2. The layouts of subsystems: (a) geothermal- and (b) solar-based for power or hydrogen generation.

combines PV and CSP technologies. This integrated system's parts work together to achieve heat recovery, hydrogen production and renewable power generation as well as thermal energy production along with fresh water supply.

The software System Advisor Model (SAM) is used to evaluate the solar sub-system based on the parameters defined and shown in Fig. 2 (b). In addition, the RETScreen software is used to evaluate the biomass production and outputs of the Combined Heat and Power (CHP) plant. The outputs are assessed based on the specific gas turbines selected for

the location. Hydrogen production is assessed based on the NEL Hydrogen A3880 electrolyser.

2.2. System environment

To analyze the proposed system, three locations around the globe were selected to better understand the needs and the renewable energy potential of different communities under variable climate conditions. This process included referring to the Global Solar Atlas (Global Solar

Atlas, 2023) to explore climate conditions around the globe. The preference was given to locations with high solar radiation potential. The global communities selected for the study are discussed. These sites were carefully chosen with consideration for geothermal potential, population density, solar radiation levels, and regional energy requirements in mind, all of which contributed to their unique renewable energy potentials. This thorough system environment evaluation ensures that the solar and geothermal resources are in each location to generate sustainable power while meeting the unique energy needs of each community.

Toowoomba, which is an Australian city [designated as A], is in the southern region of Queensland and is positioned to receive a lot of sunlight. It is the perfect place to capture solar energy because of the favorable levels of solar radiation as indicated by the Global Solar Atlas (Global Solar Atlas, 2023). As a result of the moderate population size, the area has land to be able to implement solar farms without running into complications with the existing communities. In addition, the area's electricity needs can be effectively met by a well-suited solar infrastructure and the city is close to major cities where the produced electricity can also be easily distributed. The Queensland, Australia region has also received new Geothermal Exploration permits (Zillmann and Makras, 2010). This makes this location ideal for both geothermal and solar for future projects and investments.

Another site that has been selected with a lot of solar potential is the city of Imperial in the USA [designated as B]. This city is situated in the California region in the United States. A good basis to produce solar power is provided by the region's abundance of sunshine. This is evident through the Global Solar Atlas (Global Solar Atlas, 2023) of the United States, where California is one few locations with warmer climates throughout the year. The number of people living in California suggests a high energy demand, which is met by solar infrastructure designed to specifically supply the region's electrical requirements rather than fossil fuel resources.

The city of Kisumu in Kenya [designated as C] has special qualities that make it a good fit for geothermal energy. The area has a reputation for having geothermal potential, which presents a chance to use this renewable resource to produce sustainable electricity. In addition, Kenya has warmer climates throughout the year. Kisumu in particular is shown to have higher solar intensity from the Global Solar Atlas (Global Solar Atlas, 2023) than other areas. The system design takes into account the area's substantial biomass supply and freshwater requirements, even with a comparatively lower electricity demand.

2.3. System requirements

The design of the presently developed system has a strong emphasis on the placement of geothermal and solar infrastructures to maximize power generation while optimizing land utilization. To guarantee an adequate supply of electricity to meet the energy needs of the respective communities, locations with good solar radiation and geothermal potential are selected. Table 1 summarizes the population size for the communities selected, and the corresponding yearly demands and supplies of freshwater, biomass, and electricity are also provided.

Table 1

The specifications and statistics for locations of interest.

Location	Population	Annual Electricity Demand (MW)	Annual Biomass Supply (x10 ³ tons)	Fresh Water Demand (x10 ⁶ L)
[A]	125,447	141	83	2331
[B]	181,253	257	94	3368
[C]	1,155,574	22	851	21471

3. Modelling and analysis

In all three of the locations that were chosen, the study conducts a thorough analysis of renewable energy sources. Several research approaches are used to study the particular components under evaluation. Thermodynamic and energy analysis is one approach used to evaluate the system. The System Advisor Model (SAM) is used to model both the concentrated and photovoltaic (PV) solar plants. The model defines the research variables to evaluate the results obtained, and to assess the viability and potential of the solar energy systems in the respective locations selected. Simultaneously, the software RETScreen is also utilized as a tool to estimate outputs and evaluate the other components in the system.

3.1. Geothermal plant

An in-depth geothermal subsystem analysis is carried out in this section, taking into account a variety of useful outputs, including power generation, space heating, and industrial heating. Thermodynamic principles are used to carry out a thorough assessment of the proposed geothermal system. This study takes a well-known approach by combining energy and exergy analyses to evaluate the performance of the overall system. The general energy balance equations (Dincer, 2021) given in (1) to (4) are used to drive the equations for evaluation of the geothermal sub-system. The general mass balance equation is written as follows:

$$\sum \dot{m}_{Inputs} = \sum \dot{m}_{Outputs} \quad (1)$$

The general energy balance equation is as follows:

$$\sum \dot{E}_{Inputs} = \sum \dot{E}_{Outputs} \quad (2)$$

The entropy balance equation is given as:

$$\sum \dot{E}n_{Inputs} + \dot{S}_{gen} = \sum \dot{E}n_{Outputs} \quad (3)$$

The exergy balance equation is as follows:

$$\sum \dot{E}x_{Inputs} = \sum \dot{E}x_{Outputs} + \dot{E}x_{Destruction} \quad (4)$$

The main outputs of the geothermal sub-system are re-arranged using the specific balance equations of the individual components in the geothermal sub-system derived using the given general balance equation given in (1) to (4). The outputs from the geothermal system are given as follows:

$$\dot{W}_{net} = \dot{W}_{T,F} + \dot{W}_{T,ORC} - \dot{W}_P \quad (5)$$

$$\dot{W}_{T,F} = \dot{m}_{1-3}(h_{1-3} - h_{1-6}) \quad (6)$$

$$\dot{W}_{T,ORC} = \dot{m}_{1-10}(h_{1-10} - h_{1-11}) \quad (7)$$

$$\dot{Q}_{out,IH} = \dot{m}_{1-5}(h_{1-5} - h_{1-8}) \quad (8)$$

$$\dot{Q}_{out,JH} = \dot{m}_{1-11}(h_{1-11} - h_{1-12}) \quad (9)$$

$$\dot{Q}_{HC} = \dot{Q}_{out,HCI} + \dot{Q}_{out,HCI} \quad (10)$$

The energy efficiencies of the single-generation, co-generation, and multi-generation are:

$$\eta_{singen,geo} = \frac{\dot{W}_T}{\dot{m}_{1-1}h_{1-1}} \quad (11)$$

$$\eta_{cogen,geo} = \frac{\dot{W}_T + \dot{Q}_{HW}}{\dot{m}_{1-1}h_{1-1}} \quad (12)$$

$$\eta_{\text{multigen,geo}} = \frac{\dot{W}_T + \dot{Q}_{HW} + \dot{Q}_{HC}}{\dot{m}_{1-1} h_{1-1}} \quad (13)$$

The exergy efficiencies of the single-generation, co-generation, and multi-generation are:

$$\psi_{\text{singen,geo}} = \frac{\dot{W}_T}{\dot{m}_{1-1} ex_{1-1}} \quad (14)$$

$$\psi_{\text{cogen,geo}} = \frac{\dot{W}_T + \dot{Q}_{HW} \left(1 - \frac{T_0}{T_s}\right)}{\dot{m}_{1-1} ex_{1-1}} \quad (15)$$

$$\psi_{\text{multigen,geo}} = \frac{\dot{W}_T + \dot{Q}_{HW} \left(1 - \frac{T_0}{T_s}\right) + \dot{Q}_{HC} \left(1 - \frac{T_0}{T_s}\right)}{\dot{m}_{1-1} ex_{1-1}} \quad (16)$$

3.2. Solar power generation

The software SAM is the main tool used in this research to perform analyses related to PV and CSP solar systems. The CSP system introduced in Fig. 2(b) is used as a reference to perform the analysis using SAM. This software enables detailed evaluations of the performance and feasibility of both CSP and PV solar technologies. Using the SAM model created, the results for CSP and PV solar systems are obtained for various scenarios on an annual basis. The results obtained from the model include system performance factor, and electrical power generated based on the solar data linked for each location to the SAM model. The software's flexibility allows for various system configurations which are optimized based on user-defined criteria including solar availability and technological specifications. In particular, the parameters selected in SAM for CSP are illustrated in Fig. 2.

3.3. Hydrogen generation

Excess electricity from solar and geothermal power sources is considered when analyzing hydrogen production using the NEL Hydrogen A3880 alkaline electrolysis. First, the outputs for the generation of renewable power are projected and evaluated by this study. The amount of excess power available is determined by analyzing all available data on power generation, including variations in generation brought on by weather or changes in the energy supply.

The study makes comprehensive calculations of hydrogen production based on the average power consumption of 45.5 kWh/kg using the NEL Hydrogen A3880 electrolyser specifications. The analysis involves estimating the amount of excess electricity that can be used for hydrogen production through the electrolysis process. This approach incorporates monthly electrical demands, which are determined from the annual demand, assuming constant consumption patterns throughout the year to further refine the analysis. The investigation's goal is to correlate hydrogen production capacities for different electricitital power available by taking the renewable power generation profiles into account. This methodology enables an evaluation of the electrolyzer's adaptability to varying energy requirements, thereby presenting possible situations in which hydrogen production is in line with fluctuating excess generated power available.

For the electrolysis, the reaction at the anode can be described by the balance equations as:



The reaction at the cathode is given by the energy balance equation (18):



The overall reaction of the electrolysis is given by the energy balance equation (19):



3.4. Biogas and hydrogen production from biomass

This section investigates the distribution of available biomass allotted for biogas production across the three locations taken into consideration. The amount of biomass available for each location is determined based on the amount of biomass reported in the International Energy Agency (IEA) annual reports for 2021. These approximations provide the baseline for the methods utilized to determine the amount of biogas generated. The study also offers an alternative approach to the utilization of biomass to produce hydrogen.

Recent studies have shown that at 670 °C and a steam-to-biomass ratio of 0.83, in steam biogas gasification systems can produce 230.28 ml/g of biomass (Temiz and Dincer, 2021). This result provides a foundation for estimating hydrogen production and acknowledges the potential need for additional heating to attain the required temperature.

Additionally, there have been studies that demonstrate the production of 115 m³/ton of biogas from 1 ton of biomass obtained from municipal solid waste (Zhu et al., 2010). For validation, this value is compared with biomass yield obtained by the software RETScreen which yields 117 m³/ton of biogas for municipal biowaste. For biogas production, a yield of 115 m³/ton of biogas is taken to provide a conservative estimate. This paper further investigates the possibility of using biomass exclusively to produce biogas. With a focus on biogas and hydrogen as two key applications, this expanded use of biomass in renewable energy frameworks increases the potential applications and utilization of energy products derived from biomass. Biomass is treated as single generation in both cases where either only biogas or hydrogen production is considered using the available biomass.

3.5. Combined heat and power plant

To analyze the Combined Heat and Power (CHP) plant component the software RETScreen Expert is used, which assesses the performance of renewable energy-based sources. By considering the biogas generated at each unique location and choosing suitable gas equipment suited to the plant's needs, this analysis seeks to ascertain the power and heat capacities of the CHP plant. This ensures that the turbine is not oversized. The biogas yield obtained from the available biomass specific to Toowoomba and Imperial is determined by selecting 4 and 5 ASE8 Honeywell turbines, respectively. The H15 Hitachi turbine is selected for the CHP plant in Kisumu, Kenya, which also has a significantly higher capacity than the ASE8 turbine. The RETScreen Expert optimizes the selection of gas turbines for each location. This process considers the variability in biomass availability and its direct impact on biogas production to ensure the best possible match between gas turbine capacity and available biogas.

4. Results and discussion

This section provides a thorough evaluation of the renewable energy options that are investigated for each of the three communities. This study evaluates the outcomes and implications of photovoltaic solar systems, concentrated solar power, biogas yields, hydrogen production, and geothermal plant performance at each location. The performance of geothermal plants, PV solar technologies, and concentrated solar power are the main subjects and the results obtained for each, and possible contributions to sustainable energy projects in these communities are discussed.

4.1. Geothermal sub-system results

The results obtained by using thermodynamics analysis and the Engineering Equation Software (EES) to solve the balance equations for the

geothermal system are shown in Table 2. Primary outputs are explained in the table. These include power generation and available heating capacities that are obtained to different uses, including space heating, industrial heating, and hot water supply.

A comparison of the energetic and exergetic efficiencies for the different geothermal operating modes is shown in Fig. 3. The multi-generation operation stands out for resulting in the highest levels of exergetic and energetic efficiencies. This higher efficiency is a result of the multi-generation system which considers power generation as well as all heating outputs. The co-generation mode, on the other hand, shows marginally lower efficiencies because it only takes into consideration power generation and hot water heating. Similarly, the multi-generation and co-generation modes demonstrate relatively higher efficiencies than the single-generation mode, which only uses the power produced by the geothermal plant.

The importance of multigeneration efficiency is observed, which includes power generation in addition to different heating outputs, is indicative of the full geothermal resource utilization. This system optimizes total energy production, meeting a variety of heating requirements in addition to electricity generation. On the other hand, the variations in efficiency between co-generation and single-generation modes highlight how crucial it is to take into account additional heat utilization in addition to power generation. The single-generation mode focuses primarily on power generation, resulting in slightly lower efficiencies due to the narrower scope of energy utilization, whereas co-generation accounts for hot water heating in addition to electricity production.

The results represent efficiencies based on a 25 °C ambient temperature. The exergy efficiency tends to decrease as the ambient temperature rises. Note that the energy destruction increases as a result of rising ambient temperatures, and this trend is directly related to it. Elevations in the surrounding temperature create a greater gradient that increases the dissipation of thermal energy within the system, thereby exacerbating the waste of energy. Declining exergy efficiency results from this increase in exergy destruction, which indicate a decreased capacity to transform available energy into useful work accordingly. To further increase the efficiency of the system, some of the excess heat can be further integrated with another organic Rankine cycle to produce more electrical power. This would increase the system's single-generation efficiency, as well as the multigenerational efficiency.

4.2. Solar power

Table 3 presents the load factors obtained from photovoltaic and concentrated solar power at the selected locations. In essence, the load factor is a measure of the generated power in relation to the plant's total capacity and is used to determine the efficiency of the plant. The data is noteworthy in that it shows that concentrated solar plants have a greater capacity for power generation than photovoltaic at the same locations.

The total power produced by concentrated solar power at each of the specified locations is shown in Fig. 4. Interestingly, the data shows that the location with the highest concentrated solar production is Imperial, US, which is in the California region. Furthermore, the monthly profiles of concentrated solar power generation in Kisumu, Kenya, and Toowoomba, Australia, demonstrate a similar pattern. This demonstrates that there are steady levels of power generation in these areas from month to month. Examining these variations in the profiles helps

Table 2
The geothermal sub-system outputs.

Parameter	Output (MW)
Power Generation	35.83
Space Heating	122.8
Industrial Heating	151.9
Hot water Heating	64.25

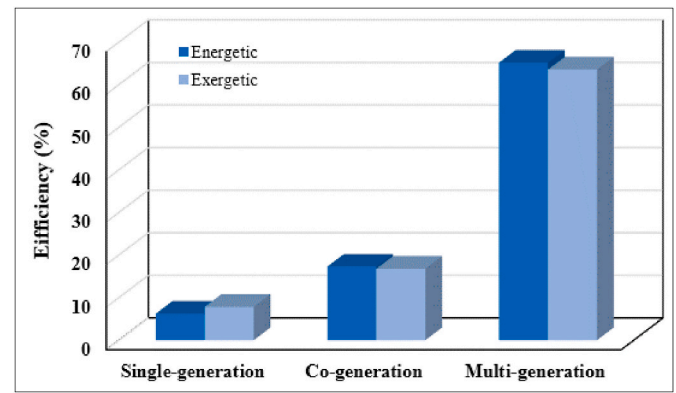


Fig. 3. The energy and exergy efficiencies for various modes of geothermal generation.

Table 3

The load factors for PV and concentrated solar power.

Location	PV Solar Load Factor	Concentrated Solar Load Factor
Toowoomba, Australia	0.117	0.194
Imperial, US	0.206	0.297
Kisumu, Kenya	0.157	0.204

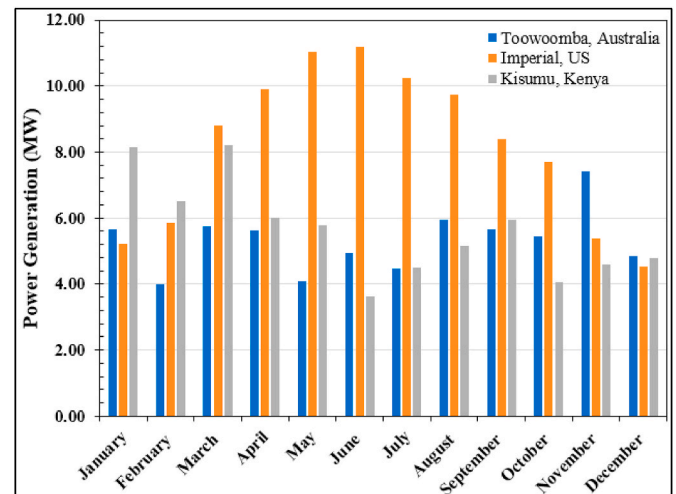


Fig. 4. The PV-based solar power generation by month for locations of interest.

determine the fluctuating CSP generation levels as well as surplus electricity, which is necessary to assess potential energy storage methods or hydrogen production opportunities.

Insights into the dynamics of power generation in each region is essential for better understanding, particularly in effectively utilizing excess renewable energy resources in various geographical contexts. Heat maps showing the power produced by CSP at the three different locations are shown in Fig. 5. Interestingly, Toowoomba, Australia experiences peak power generation from November to February, which coincides with the region's summer season. This higher power output in these months corresponds with the summertime intensified solar irradiance in Australia. Furthermore, there is an apparent increase in solar energy intensity at sunrise and sunset. It can be observed from Fig. 5 (a) that between hours of 9–19, there is very low power generated shown by the dark blue region. This indicates that there is an increase in power generation early and late hours of the day throughout the year. On the other hand, power generation is distributed rather consistently in Imperial, US. At certain times of the day, there tends to be reduced

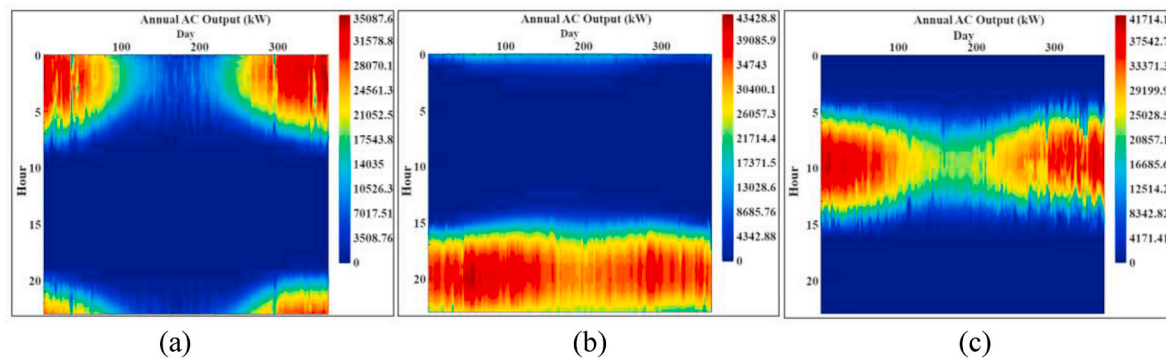


Fig. 5. The PV-based solar power heat map for locations of interest: (a) Toowoomba, Australia, (b) Imperial, US and (c) Kisumu, Kenya.

intensities of solar power, particularly later in the day. In addition, there is a noticeable increase in the intensity of solar power during the day.

Similarly, Kisumu in Kenya experiences a significant increase in power generation from November to March, which coincides with the months when the region produces its most power. During other months, there is a noticeable drop in the generation of power. The patterns of power generation that are different between these sites provide important information about how local seasons and solar irradiance affect CSP power output. Solar energy concentration at certain times of the day provides additional insights into the critical windows of increased power generation and advances in knowledge of local solar energy dynamics and possible applications.

The total power produced by the PV solar panels at the three designated locations is shown in Fig. 6. In general, the data shows that PV solar produces slightly less electricity than CSP. A notable decline in PV solar generation from May to June in Toowoomba, Australia is observe. On the other hand, PV solar generation is noticeably higher in Imperial, US, during the same period. Furthermore, the figures indicate a trend of PV solar generation that is reasonably constant between the three locations from November to January, demonstrating similar power production during this period. Nonetheless, the distinct regional variances are noticeable in various months, highlighting the impact of seasonal variations on PV solar output accordingly.

The comparison of both concentrated solar and PV solar provides better insights into both solar energy dynamics. The results show that concentrated solar power tends to have higher power generation and efficiency compared to PV. However, during certain hours, PV produces more power than concentrated solar. Using a hybrid approach allows the

communities to benefit from both solar applications when the other is less efficient. In addition, cost reductions can also be obtained since concentrated solar generally costs significantly more.

The heat maps in Fig. 7 provide useful information about the power produced by PV solar at the three designated locations. In Toowoomba, Australia, increased electricity generation is noticeable from November to February, especially early in the day. However, Kisumu, Kenya shows a rise in power generation during the summer, specifically from June to August. At these locations, the concentration of solar power seems to be stronger in the midday hours. The distribution of power generation in Imperial, US, is rather consistent throughout the day. Nonetheless, there are times when the output of PV solar power is relatively lower. Summertime power generation is higher, which corresponds with higher solar radiation values and more hours of available sunlight.

4.3. Hydrogen production estimate for electrolysis

A detailed representation of the outputs from geothermal, photovoltaic, and concentrated solar power is shown in Fig. 8, along with the corresponding hydrogen production from excess electricity. This illustration provides a comprehensive overview of the energy outputs from various renewable sources and the potential for hydrogen production in Toowoomba, Australia in Fig. 8a, Imperial, US in Fig. 8b, and Kisumu, Kenya in Fig. 8c.

The fluctuation between the solar power generated at the different locations is seen. There is also a variation in the amount solar power generated from PV and CSP. Relatively, the power produced from CSP is slightly higher. In comparison to solar, geothermal has higher capacities for electrical power production. Integrating geothermal power into the system increases the yield for hydrogen production as well as the amount of power that is available for the communities. These results emphasize the benefits of incorporating a combination of renewables which are better able to meet the energy needs of the community.

Furthermore, Fig. 9 represents the highest possible hydrogen yield that can be produced at the designated locations by utilizing all of the excess renewable power generated. The estimated yearly hydrogen yields for Kisumu, Kenya; Toowoomba, Australia; and Imperial, United States are 235 tons, 216 tons, and 313 tons, respectively.

Evaluating these results obviously shows different trends for different places, which correspond to different renewable energy potentials. Kisumu, Kenya exhibits greater hydrogen potential, which can be attributed to particular climatic conditions that promote greater renewable power generation. The Imperial, US, has a marginally lower potential for producing hydrogen than other locations. Furthermore, the illustration of hydrogen production from excess electricity from Fig. 8, emphasizes the complex interactions among climate conditions, renewable power generation, and electrical demands. The higher solar radiation intensity is associated with a tendency toward higher hydrogen production. Interestingly, different electrical demands have a major influence on the availability of renewable power for the

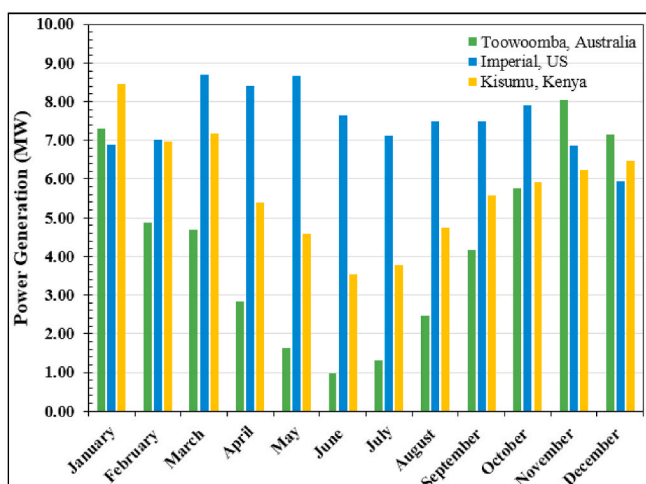


Fig. 6. The concentrated solar power generation by month for locations of interest.

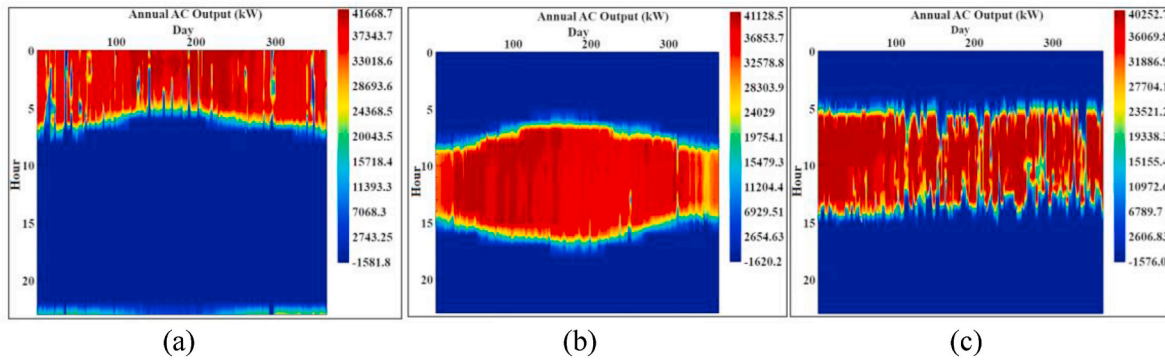


Fig. 7. The concentrated solar power generation for locations of interest: (a) Toowoomba, Australia, (b) Imperial, US and (c) Kisumu, Kenya.

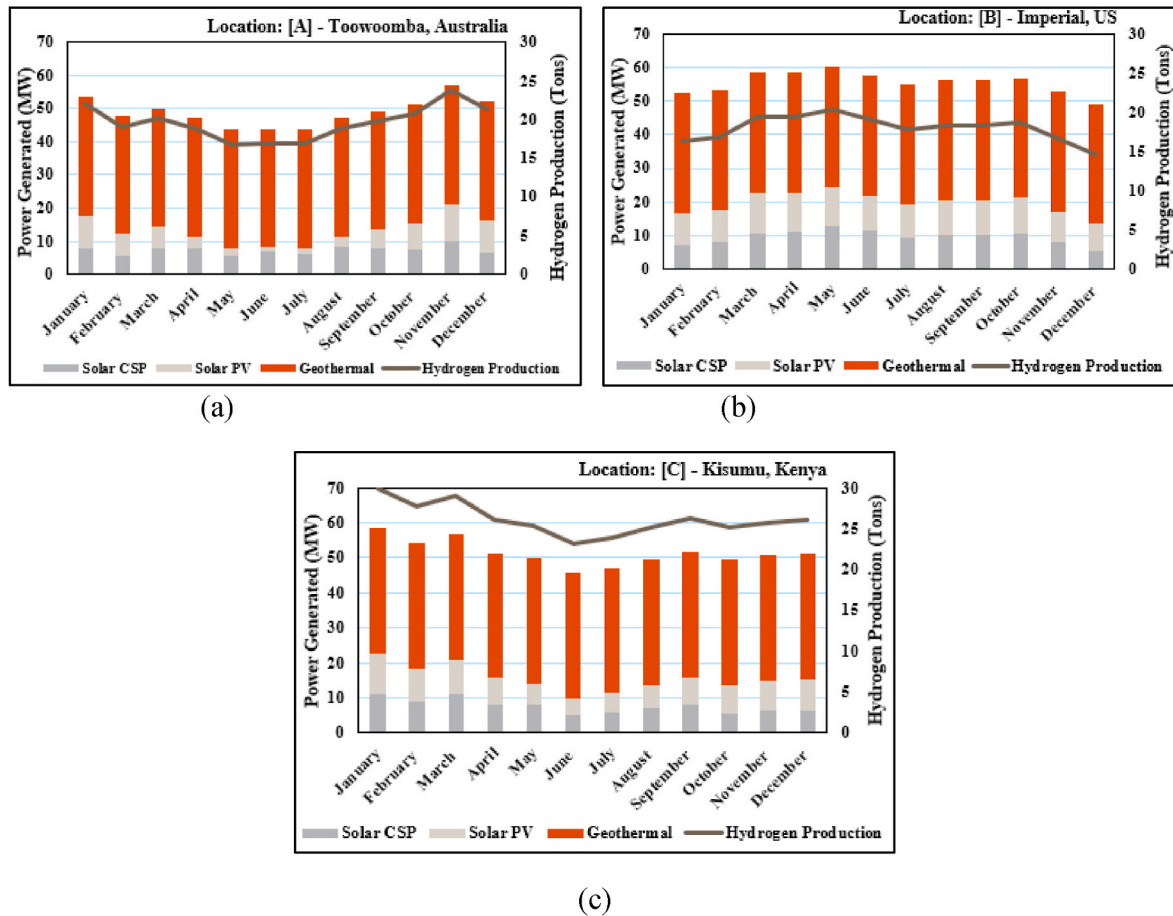


Fig. 8. The solar and geothermal power and hydrogen production based on excess renewable power available for (a) Toowoomba, Australia (b) Imperial, US and (c) Kisumu, Kenya.

production of hydrogen, which is consistent with earlier findings.

4.4. Biogas and hydrogen production from biomass

The biogas and hydrogen production yields are estimated based on the biomass resources available at the selected locations and are listed in Table 4. These are estimated based on the total amount of biomass utilized for each source, assuming the biomass is not used for hydrogen production when biogas is produced and vice versa.

The estimated amount of biogas that Toowoomba, Australia can produce from its available biomass of municipal solid waste is $9.44 \times 10^6 \text{ m}^3$. In contrast, Imperial, US expects to produce slightly more

biogas, roughly $10.8 \times 10^6 \text{ m}^3$. Surprisingly, Kisumu, Kenya, has a significantly larger biogas potential $97.8 \times 10^6 \text{ m}^3$, according to the results. These differences in biogas output indicate a significant variation in the amount of biomass from municipal solid waste that is available in each of these locations, which affects the potential for each location to produce biogas. In terms of producing hydrogen from solid biomass, Toowoomba, Australia is projected to produce about 17 tons of hydrogen. A somewhat higher potential for hydrogen production, roughly 20 tons, is estimated by Imperial, US. Kisumu, Kenya, on the other hand, anticipates a noticeably higher hydrogen yield of about 178 tons. These estimates demonstrate the unique capacities of these sites for producing hydrogen from solid biomass resources, highlighting Kisumu,

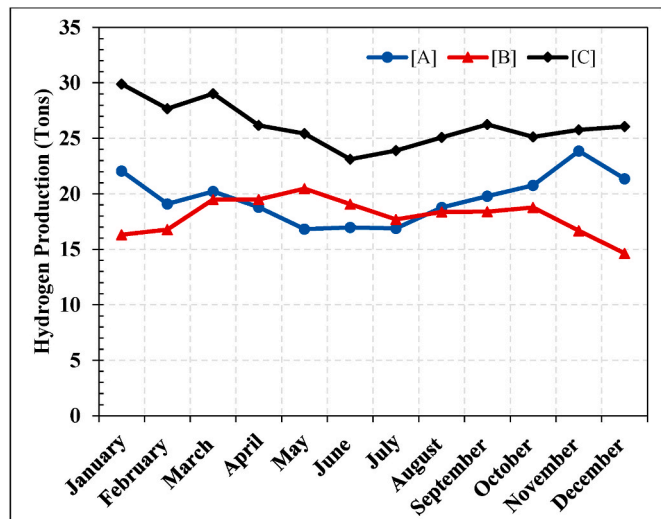


Fig. 9. The hydrogen production potential based on total renewable power available for locations of [A], [B], and [C].

Table 4
The biogas and hydrogen production yields.

Location	Biogas produced from available municipal solid waste biomass ($\times 10^6 \text{ m}^3$)	Hydrogen produced from available solid biomass (tons)
Toowoomba, Australia	9.44	17
Imperial, US	10.8	20
Kisumu, Kenya	97.8	178

Kenya as a prominent location for green hydrogen production.

The differences in the location's estimates of biogas and hydrogen production highlight how differently these areas can use the biomass resources that are available to them. These estimates also demonstrate Kisumu, Kenya's significant potential in this area and provide important insights into the possible contributions of biomass-based energy resources to regional energy initiatives. Biomass utilization is treated as single generation in each case. This is assuming the biomass is not used for hydrogen production when biogas is produced and vice versa. From the estimates, there is potential for producing more biogas than hydrogen since there are higher yields obtained. However, for communities where there is expected to be a higher demand for hydrogen than biogas, then the biomass should be utilized for hydrogen production.

4.5. Combined heat and power plant outputs

Table 5 shows the Combined Heat and Power (CHP) plant's power and heat recovery capabilities at different locations, as determined by RETScreen Expert. These calculations are based exclusively on the

Table 5
The power and heat generation from CHP plant for the biogas available.

Location	Toowoomba, Australia	Imperial County, US	Kisumu County, Kenya
Fuel Consumption (m^3)	8,760,000	8,760,000	6,132,000
Power Capacity (MW)	2.4	2.4	16.6
Heating Capacity (MW)	0.89	0.89	6.2

biogas production data shown in Table 4. The recovered heat meets the critical requirement for process heating by supporting essential industrial processes. The potential for greater biogas production and the corresponding capacities for power and heat recovery within the CHP plant are correlated across all of the locations evaluated. This correlation is mainly explained by the fact that biogas is now more readily available, which permits the CHP plant to use it more extensively.

Kisumu County uses a slightly smaller volume of $6,132,000 \text{ m}^3$, whereas Toowoomba and Imperial County show similar consumption of $8,760,000 \text{ m}^3$. Regarding the power generation capacity, the power generation capacity of the CHP plant varies significantly between these sites. Kisumu County has a significantly larger power capacity of 16.6 MW than Toowoomba or Imperial County, both of which have similar power capacities of 2.4 MW. This notable discrepancy suggests different power output capacities related to biogas use.

These results highlight how the energy production capacities of CHP plant vary depending on the region and the amount of biogas consumed. When compared to Toowoomba and Imperial County, Kisumu County has substantially greater power and heating capacities, even though it uses less biogas. Compared to the other locations, the data indicates that Kisumu County's CHP plant exhibits superior biomass resources, allowing for increased power and heating outputs per unit of biogas consumed. Comprehending these capacities is pivotal to optimizing energy production and utilization strategies customized to the unique capacities and demands of each region considered.

5. Conclusions

The need for sustainable and clean energy solutions has drastically increased due to initiatives to reduce carbon emissions. In this study, a novel system is evaluated that utilizes a combination of renewable resources to produce a variety of useful outputs, including electricity, hydrogen, space heating, industrial heating, fresh water, hot water, and biogas. Three selected locations were the focus of the investigation, and each was analyzed to determine its solar potential, geothermal power potential, and potential for producing hydrogen using renewable sources. The solar plant in California, USA, showed off its substantial solar potential by demonstrating high solar outputs. A commendable solar potential was also demonstrated by Kenyan and Australian communities. By incorporating a wide range of renewable sources, including solar and geothermal energies, for hydrogen production, this study demonstrates the accessibility of multiple energy outputs.

The highlights of this study are as follows:

- The load factor of 29% concentrated solar power generation for Imperial, United States, resulted in the highest solar power efficiency.
- The overall energy and exergy efficiencies of the geothermal plant are obtained as 65.15% and 63.54% respectively.
- Kisumu, Kenya demonstrated a high biogas yield, roughly $97.8 \times 10^6 \text{ m}^3$, which made a substantial contribution to the city's overall energy portfolio.
- The annual estimates for hydrogen generation using solar and geothermal sources were estimated as 235 tons, 216 tons, and 313 tons for Toowoomba, Australia, Imperial, United States, and Kisumu, Kenya respectively.

Although solar power production in California, USA, is reported to be particularly strong, Australia and Kenya also have promising solar potential, though their capacity to produce hydrogen is somewhat limited because of their higher energy needs. The conclusions draw attention to the interplay among the capacities of these locations for producing hydrogen, energy demands, and renewable energy potential. To meet the various energy demands of the communities under consideration, there is an emphasis on the necessity of customized strategies to effectively harness solar and geothermal energies, ensuring a balance

between energy generation and hydrogen production. It is important to consider, that some energy sources are considered alternative sources to other modes of operation. Although the study considers outputs from all sources, the energy source is recommended to be selected based on the needs of the specific community.

This paper advances the current knowledge and understanding for sustainable energy by providing a comprehensive combination of renewable energy sources which can be utilized to produce multiple outputs. PV solar, concentrated solar, and geothermal are evaluated in three distinct cities located in Australia, United States and Kenya. In particular, these locations have not been studied for these combinations of energy sources in previous work. Furthermore, the study provides greater insights into power production from PV and concentrated solar sources and the differences associated with these energy sources. The study also highlights the importance of the energy dynamic balance through different modes of electricity and hydrogen production for storage. The study recommends some suggestions for further expansion of the work. An economic assessment is suggested to consider the costs and the acceptance of certain energy sources for the different communities. In addition, the environmental impacts and long-term feasibility

of the system can be considered. An experimental study can also be carried out to obtain the actual solar, geothermal, and hydrogen being provided at the various locations for different periods of the year.

CRediT authorship contribution statement

Muarij Khalil: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ibrahim Dincer:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Some data may be made available on request.

Nomenclature

CO_2	Carbon Dioxide
ex	Specific exergy (kJ/kg)
$\dot{E}x$	Exergy rate (kW)
$\dot{E}x_D$	Exergy destruction rate (kW)
$\dot{E}x^Q$	Exergy rate of heat transfer (kW)
h	Specific enthalpy (kJ/kg)
H_2	Hydrogen
LF	Load Factor
$\dot{m}$	Mass flow (kg/s)
$\dot{m}_{in}$	Mass flow input (kg/s)
$\dot{m}_{out}$	Mass flow output (kg/s)
P	Pressure (kPa)
$\dot{Q}$	Heat transfer rate (kW)
s	Specific entropy (kJ/kgK)
s_0	Specific entropy at reference conditions (kJ/kgK)
$\dot{S}$	Entropy rate (kJ/kg/sK)
T	Temperature (K)
T_0	Ambient temperature (K)

Greek Letters

η	Energy efficiency (%)
ψ	Exergy efficiency (%)

Acronyms

BCS	Biogas Collection and Storage
BG	Biogas
BPF	Biogas Production Facility
CHP	Combined Heat and Power
EES	Engineering Equation Solver
EV	Electric Vehicle
EVS	Electric Vehicle Station
HGP	Hydrogen Generation Plant
PEM	Proton Exchange Membrane
PV	Photovoltaics
WTP	Water Treatment Plant
WWTP	Wastewater Treatment Plant
SAM	System Advisor Model

Appendix

Table A1
The geothermal component balance equations

Mass Balance	Energy Balance	Entropy Balance	Exergy Balance
Throttle valve			
$\dot{m}_{1-1} = \dot{m}_{1-2}$	$\dot{m}_{1-1}h_{1-1} = \dot{m}_{1-2}h_{1-2}$	$\dot{m}_{1-1}s_{1-1} + S_{gen,TV} = \dot{m}_{1-2}s_{1-2}$	$\dot{m}_{1-1}ex_{1-1} = \dot{m}_{1-2}ex_{1-2} + \dot{Ex}_{d,TV}$
Separator			
$\dot{m}_{1-2} = \dot{m}_{1-3} + \dot{m}_{1-4}$	$\dot{m}_{1-2}h_{1-2} = \dot{m}_{1-3}h_{1-3} + \dot{m}_{1-4}h_{1-4}$	$\dot{m}_{1-2}s_{1-2} + S_{gen,Sep} = \dot{m}_{1-3}s_{1-3} + \dot{m}_{1-4}s_{1-4}$	$\dot{m}_{1-2}ex_{1-2} = \dot{m}_{1-3}ex_{1-3} + \dot{m}_{1-4}ex_{1-4} + \dot{Ex}_{d,Sep}$
Turbine			
$\dot{m}_{1-3} = \dot{m}_{1-6}$	$\dot{m}_{1-3}h_{1-3} = \dot{m}_{1-6}h_{1-6} + \dot{W}_{T,F}$	$\dot{m}_{1-3}s_{1-3} + S_{gen,T,F} = \dot{m}_{1-6}s_{1-6}$	$\dot{m}_{1-3}ex_{1-3} = \dot{m}_{1-6}ex_{1-6} + \dot{Ex}_{d,T,F}$
Heat Exchanger HW			
$\dot{m}_{1-6} = \dot{m}_{1-7}$	$\dot{m}_{1-6}h_{1-6} = \dot{m}_{1-7}h_{1-7} + \dot{Q}_{out,HW}$	$\dot{m}_{1-6}s_{1-6} + S_{gen,HEI} = \dot{m}_{1-7}s_{1-7} + \frac{\dot{Q}_{out,HW}}{T}$	$\dot{m}_{1-6}ex_{1-6} = \dot{m}_{1-7}ex_{1-7} + \dot{Q}_{out,HW}\left(1 - \frac{T_0}{T}\right) + \dot{Ex}_{d,HEI}$
Heat Exchanger HI			
$\dot{m}_{1-5} = \dot{m}_{1-8}$	$\dot{m}_{1-5}h_{1-5} = \dot{m}_{1-8}h_{1-8} + \dot{Q}_{out,SH}$	$\dot{m}_{1-5}s_{1-5} + S_{gen,HEII} = \dot{m}_{1-8}s_{1-8} + \frac{\dot{Q}_{out,SH}}{T}$	$\dot{m}_{1-5}ex_{1-5} = \dot{m}_{1-8}ex_{1-8} + \dot{Q}_{out,SH}\left(1 - \frac{T_0}{T}\right) + \dot{Ex}_{d,HEII}$
Evaporator			
$\dot{m}_{1-4} = \dot{m}_{1-5}, \dot{m}_{1-9} = \dot{m}_{1-10}$	$\dot{m}_{1-4}h_{1-4} + \dot{m}_{1-9}h_{1-9} = \dot{m}_{1-5}h_{1-5} + \dot{m}_{1-10}h_{1-10}$	$\dot{m}_{1-4}s_{1-4} + \dot{m}_{1-9}s_{1-9} + S_{gen,Eva} = \dot{m}_{1-5}s_{1-5} + \dot{m}_{1-10}s_{1-10}$	$\dot{m}_{1-4}ex_{1-4} + \dot{m}_{1-9}ex_{1-9} = \dot{m}_{1-5}ex_{1-5} + \dot{m}_{1-10}ex_{1-10} + \dot{Ex}_{d,Eva}$
ORC, Turbine			
$\dot{m}_{1-10} = \dot{m}_{1-11}$	$\dot{m}_{1-10}h_{1-10} = \dot{m}_{1-11}h_{1-11} + \dot{W}_{T,ORC}$	$\dot{m}_{1-10}s_{1-10} + S_{gen,T,ORC} = \dot{m}_{1-11}s_{1-11}$	$\dot{m}_{1-10}ex_{1-10} = \dot{m}_{1-11}ex_{1-11} + \dot{Ex}_{d,T,ORC}$
Pump			
$\dot{m}_{1-12} = \dot{m}_{1-9}$	$\dot{m}_{1-12}h_{1-12} + \dot{W}_P = \dot{m}_{1-9}h_{1-9}$	$\dot{m}_{1-12}s_{1-12} + S_{gen,P} = \dot{m}_{1-9}s_{1-9}$	$\dot{m}_{1-12}ex_{1-12} = \dot{m}_{1-9}ex_{1-9} + \dot{Ex}_{d,P}$
Heat Exchanger HIII			
$\dot{m}_{1-11} = \dot{m}_{1-12}$	$\dot{m}_{1-11}h_{1-11} = \dot{m}_{1-12}h_{1-12} + \dot{Q}_{out,HI}$	$\dot{m}_{1-11}s_{1-11} + S_{gen,HEIII} = \dot{m}_{1-12}s_{1-12} + \frac{\dot{Q}_{out,HI}}{T}$	$\dot{m}_{1-11}ex_{1-11} = \dot{m}_{1-12}ex_{1-12} + \dot{Q}_{out,HI}\left(1 - \frac{T_0}{T}\right) + \dot{Ex}_{d,HEIII}$

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